

Covered: Orthogonal Projections are Hermitian, Spectral Theorem for Hermitian Operators, Any Operator/Matrix is Unitarily Congruent to Triangular Operator/Matrix

Proposition 0.1. *Let $P \in L(V)$. Then P is an orthogonal projection if and only $P^2 = P$ and $P^* = P$*

Proof. ($\Rightarrow$) Suppose P is orthogonal projection, then $P^2 = P$ and $\text{Range}(P) \perp \ker(P)$. Let $W = \text{Range}(P)$. Then $W^\perp = \ker(P)$, giving $V = W \oplus W^\perp$. For any $x, y \in V$, then we have the following representation,
 $x = x_1 + x_2$, $x_1 \in W$, $x_2 \in W^\perp$ and $y = y_1 + y_2$, $y_1 \in W$, $y_2 \in W^\perp$.
Then $P(x) = x_1$ and $P(y) = y_1$. Moreover,

$$\langle P(x), y \rangle = \langle x_1, y_1 + y_2 \rangle = \langle x_1, y_1 \rangle + \langle x_1, y_2 \rangle = \langle x_1, y_1 \rangle$$

$$\langle x, P(y) \rangle = \langle x_1 + x_2, y_1 \rangle = \langle x_1, y_1 \rangle + \langle x_2, y_1 \rangle = \langle x_1, y_1 \rangle$$

Therefore, $\langle P(x), y \rangle = \langle x, P(y) \rangle$ for all $x, y \in V$, by uniqueness of adjoint of P , we have, $P^* = P$.

($\Leftarrow$) Suppose $P^2 = P$ and $P^* = P$. Let $W = \text{Range}(P)$. Any vector $x \in V$ can be decomposed as, $x = Px + (x - Px)$.
Note that $P(x - Px) = Px - P^2x = Px - Px = 0$, which means $x - Px \in \ker(P)$.
For $x \in V$,

$$\langle Px, x - Px \rangle = \langle x, P^*(x - Px) \rangle = \langle x, P(x - Px) \rangle = \langle x, 0 \rangle = 0$$

Thus, $\text{Range}(P) \perp \ker(P)$, and P is an orthogonal projection. $\square$

Theorem 0.2 (Spectral Theorem). *Let V be a finite-dimensional inner product space over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$) and let $T \in L(V)$. The following statements are equivalent:*

1. T is Hermitian ($T^* = T$).
2. There exists an orthonormal basis $\mathcal{B}$ of V consisting of eigenvectors of T with real eigenvalues.
3. There exists an orthonormal basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}}$ is a real diagonal matrix.
4. All eigenvalues of T are real and $V = W_1 \oplus W_2 \oplus \dots \oplus W_k$ (orthogonal direct sum), where $W_i \perp W_j$ for $i \neq j$, and W_i is the eigenspace corresponding to distinct eigenvalue λ_i .
5. $T = \sum_{i=1}^k \lambda_i P_i$, where $\lambda_i \in \mathbb{R}$ are distinct eigenvalues and P_i are orthogonal projections satisfying, $\sum_{i=1}^k P_i = I$ and $P_i P_j = \delta_{ij} P_j$

Proof. (i) $\Rightarrow$ (ii) and (ii) $\Rightarrow$ (iii) is already proved

(iii) $\Rightarrow$ (iv) is immediate.

(iv) $\Rightarrow$ (v) Let P_i be the projection operator onto W_i associated with $V = W_1 \oplus W_2 \oplus \cdots \oplus W_k$. Then $\sum_{i=1}^k P_i = I$, $T = \sum_{i=1}^k \lambda_i P_i$, and $P_i P_j = \delta_{ij} P_j$.

To show each P_j is an orthogonal projection, we show $\text{Range}(P_j) \perp \ker(P_j)$: Let $v \in V$ and $w \in \ker(P_j) = W_1 \oplus W_2 \oplus \cdots \oplus W_{i-1} \oplus W_{i+1} \oplus \cdots \oplus W_k$. Write $v = \sum_{i=1}^k v_i$, where $v_i \in W_i$ and $w = \sum_{i \neq j} \alpha_i v_i$, for some $\alpha_i \in \mathbb{F}$. Then,

$$\langle P_j v, w \rangle = \left\langle v_j, \sum_{i \neq j} \alpha_i v_i \right\rangle = \sum_{i \neq j} \bar{\alpha}_i \langle v_j, v_i \rangle = 0$$

since $W_j \perp W_i$ for all $i \neq j$. Hence, $\text{Range}(P_j) \perp \ker(P_j)$. $\square$

(v) $\Rightarrow$ (i). Suppose $T = \sum_{i=1}^k \lambda_i P_i$, where $\lambda_i \in \mathbb{R}$ and $P_i^* = P_i$. Taking the adjoint:

$$T^* = \left(\sum_{i=1}^k \lambda_i P_i \right)^* = \sum_{i=1}^k \bar{\lambda}_i P_i^*$$

Since $\lambda_i \in \mathbb{R}$ ($\bar{\lambda}_i = \lambda_i$) and $P_i^* = P_i$, we obtain:

$$T^* = \sum_{i=1}^k \lambda_i P_i = T$$

Thus, T is Hermitian

Hence they are equivalent. $\square$

Corollary 0.3 (Matrix Version). A matrix $A \in M_n(\mathbb{F})$ is Hermitian ($A^* = A$) if and only if there exists a unitary matrix U such that:

$$U^* A U = D$$

where D is a real diagonal matrix.

Remark 0.4 (Diagonalizability vs Self-Adjointness). Question: Above theorem raises question that if $T \in L(V)$ is diagonalizable with all real eigenvalues. Is T necessarily Hermitian?

Answer: No. An operator T is diagonalizable with real eigenvalues if and only if T is **similar** to a Hermitian operator. It is Hermitian itself if and only if it is **unitarily diagonalizable**.

Take $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined as $T(x, y) = (x + y, y)$, it's corresponding matrix representation with respect to standard basis is, $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$.

Then T is diagonalizable but not Hermitian.

Theorem 0.5. Let V be a finite-dimensional complex inner product space, and let $T \in L(V)$. Then there exists an orthonormal basis $\mathcal{B}$ of V such that the matrix representation $[T]_{\mathcal{B}}$ is upper triangular, with its diagonal entries equal to the eigenvalues of T .

Proof. We proceed by induction on $n = \dim V$.

If $\dim V = 1$, any 1×1 matrix representation is trivially upper triangular.

Assume the result holds for all complex inner product spaces of dimension less than n .

Inductive Step ($\dim V = n$): Since V is a complex vector space, T has at least one eigenvalue $\lambda_1 \in \mathbb{C}$. Let x_1 be a corresponding unit eigenvector, so that, $Tx_1 = \lambda_1 x_1$ and $\|x_1\| = 1$.

Define $W = \text{span}\{x_1\}$. Then $V = W \oplus W^\perp$, with $\dim(W^\perp) = n - 1$.

Let $P : V \rightarrow W^\perp$ be the orthogonal projection onto $W^\perp$. Consider the restricted operator:

$$PT|_{W^\perp} : W^\perp \rightarrow W^\perp$$

Since $\dim(W^\perp) = n - 1$, by the induction hypothesis, there exists an orthonormal basis $\mathcal{B}_1 = \{x_2, \dots, x_n\}$ of $W^\perp$ such that the matrix representation:

$$[PT|_{W^\perp}]_{\mathcal{B}_1} = D_1$$

is upper triangular.

Now define $\mathcal{B} = \{x_1, x_2, \dots, x_n\}$. Since $x_1 \in W$, $x_1 \perp W^\perp$, and $\mathcal{B}_1$ is an orthonormal basis for $W^\perp$, it follows that $\mathcal{B}$ is an orthonormal basis of V .

$$\begin{aligned} Tx_j &= PT|_{W^\perp}(x_j) + (I - PT|_{W^\perp})(x_j) \\ &= \sum_{i=2}^n \alpha_{ji} x_i + \alpha_{j1} x_1 \\ Tx_j &= \sum_{i=1}^n \alpha_{ji} x_i \end{aligned}$$

$[T]_{\mathcal{B}}$ is upper triangular. This completes the proof. $\square$

Remark 0.6. In above proof, we can also prove the latter arguments using matrices, here is the proof:

With respect to $\mathcal{B}$, since $Tx_1 = \lambda_1 x_1$, the matrix representation of T has the form:

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & * \\ 0 & D_1 \end{pmatrix}$$

Since D_1 is upper triangular, $[T]_{\mathcal{B}}$ is also upper triangular, with the diagonal entries containing the eigenvalues of T . This completes the proof.

Theorem 0.7. Let $A \in M_n(\mathbb{C})$ be any $n \times n$ complex matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (with multiplicity). Then there exists a complex unitary matrix U such that:

$$U^*AU = \Delta$$

where Δ is an upper triangular matrix whose diagonal entries are precisely the

eigenvalues of A :

$$\Delta = \begin{pmatrix} \lambda_1 & * & \dots & * \\ 0 & \lambda_2 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$$